Design and Analysis of Algorithms

Chapter One

Introduction

Design and Analysis of Algorithm: Objectives

This subject focus on following points

- To Construct the algorithms for the problems to be solved or already solved by other methods, the problems may be from diverse fields and not restricted to Computer Science only.
- To prove that a proposed algorithm solve the problem correctly.
- To analyze the time and space requirements of an algorithm in **standard**, **asymptotical notations**, that is independent of particular machine.
- To prove that the **proposed algorithm solve** the problem **faster** than other solutions.

Introduction

- What is an Algorithm
  - Definition, characteristics, Writing an algorithm
  - Algorithm and Basic Concept
- Algorithm Complexity
  - Analysis of the algorithms
  - Computing run time of algorithms
  - Bounds
- Important Problem Type
  - Searching and Sorting- linear and binary search
  - String processing
- Fundamental Data Structure
  - Liner data Structure :- stack and queue
  - Non Liner Data structure :- tree and graph

Algorithm: Definition

An **algorithm** is a **steps of operations to solve a problem performing calculation, data processing**, and **automated reasoning** tasks. And it is an **efficient method** that can be **expressed within finite** amount of **time** and **space**.

- It can have **zero** or **more inputs.**
- It must have **at least one output.**
- It should be **efficient** both in terms of **memory** and **time**.
- It should be **finite.**
- Every statement should be **unambiguous.**

Algorithm is **independent** from **any programming language.**

Note: A program can run **infinitely**, for example a server program it runs **24X7X365**. But an algorithm must be **finite**.

Characteristics of an Algorithm

Not all procedures can be called **an algorithm**.

An algorithm should have the following characteristics

- **Unambiguous** – it should be **clear** and **unambiguous**.

Each of its steps (or phases), and their inputs/outputs should be clear and must lead to only one meaning.

- **Input** – An algorithm should **have 0** or **more** well-defined inputs.
- **Output** – An algorithm should have **1** or **more** well-defined outputs, and should match the desired output.
- **Finiteness** – it must **terminate** after a finite number of steps.
- **Feasibility** – Should be **feasible** with the **available resources**.
- **Independent** – it should be **independent** of any programming code.

Note: It must have a termination criteria

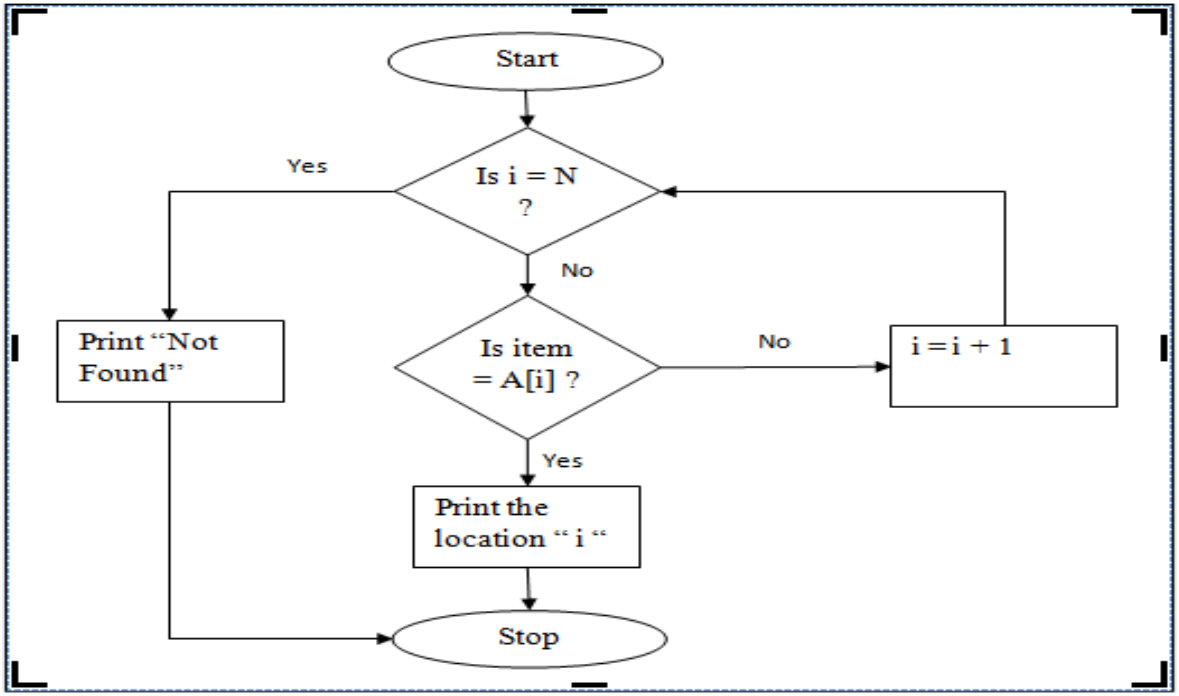
Writing an Algorithm

An algorithm can be **written** in following ways:

- **Natural Language Representation:** Use of **Natural language in writing** and **algorithm** can be ambiguous and therefore algorithm may lack the characteristic of being **definite**.
- **Flowcharts: Graphical representation** of algorithmic steps, flowcharts are **not suitable** to write the solutions for **complex problems**.
- **Pseudocode: an informal language, combination of human language and matimatical expression.**
- it has an advantage of **being easily converted** into any programming language. A Table of pseudocode conventions are given in textbook. Mostly used.

Flowchart

- Flowcharts **pictorially depict** a process.
- They are **easy to understand** and are **commonly used** in case of **simple problems**.



Pseudo Code

- The **pseudo code** has an advantage of **being easily converted** into any programming language.
- This way of writing algorithm is **most acceptable** and **most widely used**.
- In order to be able to write a pseudo code, one must be familiar with the conventions of writing it.

```
Algorithm LinearSearch(A, n,  item)
{
    for i := 1 to n step 1 do
    {
        if(A[i] = item) then
        {
            write i;
            exit;
        }
    }
    write "Not Found"
}
```

1.Write an algorhim that find sum of N positive integer numbers.

2. An algorhim find the biggest among N numbers.

Algorithm design strategies

- **Brute force**
  - **it is an approach** that **finds all the possible solutions** to **find a satisfactory solution** to a given problem
- **Divide and conquer**
  - **Breaking down** a problem into **two** or **more** sub-problems of the **same (or related) type**, until these become **simple enough** to be **solved directly**.
  - The **solutions** to the sub-problems are then **combined** to give a solution to the **original problem.**
- **Decrease and conquer** **(instance ->An occurrence of something)**
  - Change an **instance** into **one smaller instance** of the **problem**.
  - **Solve** the **smaller instance**.
  - **Convert** the solution of the **smaller instance** into a solution for the **larger instance**.

Algorithm design strategies

- **Transform and conquer**
  - Transform problem to **easiar/familar form** or **use another represenation** or **another porblem instance** then solve it.
- **Greedy approach**
  - Used to solve **optimization** problem.(**optimization**:-problem which required **max** or **min** result)
  - we follow **predifined procedure** to get the **potimal result**
  - The **problem could be** solved in **iterations.**
- **Dynamic programming**
  - Used to solve **optimization** problem.
  - find **all posible feasible solution** and pickup the best solution.
  - **Time consuming**
  - most of the time solved by **recursion formual**

Algorithm: Design Considerations

The **five most essential things** are to **be considered** while **writing an algorithm** are as follows:

- Time taken
- Memory usage
- Input
- Process
- Output

Algorithm Complexity

Suppose **X** is an **algorithm** and **n** is the size of **input data**, the **time** and **space** used by the **algorithm X** are the two main factors, which decide the **efficiency** of X.

- **Time Factor − Time is measured** by **counting** the **number of key operations** such as comparisons in the sorting algorithm.

- **Space Factor − Space is measured** by **counting** the maximum memory space required by the **algorithm**.

The complexity of an algorithm **f(n)** gives the **running time** and/or **the storage space** required by the **algorithm** in terms of **n** as the size of input data.

Types of the Algorithm Analysis

Efficiency of an algorithm can be **analyzed** at **two different stages**, **before implementation** and **after implementation**. They are the following

- **A Priori Analysis** – This is a **theoretical analysis** of an algorithm.
  - **Efficiency of an algorithm** is measured **by assuming** that all **other factors**, for example, processor speed, are **constant** and **have no effect** on the implementation.
- **A Posterior Analysis** – This is **an empirical analysis** of an algorithm.
  - The **selected algorithm** is implemented **using programming language**. This is then executed on target computer machine.
  - In this analysis, **actual statistics** like **running time** and **space required**, are collected.

Why to perform analysis of the Algorithms

The analysis of an algorithm is done because:

- The **analysis of algorithm** can be **more reliable** than **experiments**.
- **Experiments can guarantee** the behavior of algorithms on **certain test cases**, **while theoretical analysis** and establishment of run time bounds can **guarantee** the **behavior of algorithm on whole domain**.
- **Analysis** helps us **to chose from many possible solutions**.
- **Performance** of the **program can be predicted** before it is **implemented**.
- By analysis we can idea of **slow** and **faster parts** of the **algorithm**, and accordingly we can plan our implementation strategies.

Analysis of algorithms

- **Issues**:
  - Correctness
  - Space efficiency
  - Time efficiency
  - Optimality
- **Approaches**:
  - Theoretical analysis
  - Empirical analysis

*Apriori analysis vs. posterior analysis

Worst-Case/ Best-Case/ Average-Case Analysis

- ***Worst-Case Analysis*** –The maximum amount of time that an **algorithm require** to **solve a problem** of **size n**.
- ***Best-Case Analysis*** –The minimum amount of time that an **algorithm require** to s**olve a problem** of **size n**. The best case behavior of an algorithm is **NOT so useful.**
- ***Average-Case Analysis*** –The average amount of time that an **algorithm require** to **solve a problem** of size n.
  - **Worst-case analysis** is more common than average-case analysis.

Asymptotic Analysis

- **Asymptotic analysis** of an **algorithm** refers to defining the **mathematical boundation/framing** of **its run-time performance**.
- **Using asymptotic analysis**, we can very well conclude the **best case**, **average case**, and **worst case** scenario of **an algorithm**.
- **Asymptotic analysis** is **input bound** i.e., if there's **no input to the algorithm**, it is **concluded** to work in a **constant time**.
- Other than the "input" **all other factors** are considered **constant**.
- **Asymptotic analysis** refers to **computing the running time** of **any operation** in mathematical units of computation.

Asymptotic Analysis

Common order-of-growth classifications

Definition. If $f(N) \sim c\, g(N)$ for some constant $c > 0$, then the order of growth of $f(N)$ is $g(N)$.

- Ignores leading coefficient.
- Ignores lower-order terms.

Ex. The order of growth of the running time of this code is N^3.

```
int count = 0;
for (int i = 0; i < N; i++)
   for (int j = i+1; j < N; j++)
      for (int k = j+1; k < N; k++)
         if (a[i] + a[j] + a[k] == 0)
            count++;
```

Typical usage. With running times.

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Asymptotic Notations

Following are the **commonly used asymptotic notations** to calculate the **running time complexity** of an **algorithm**.

- ✓ O Notation
- ✓ Ω Notation
- ✓ θ Notation

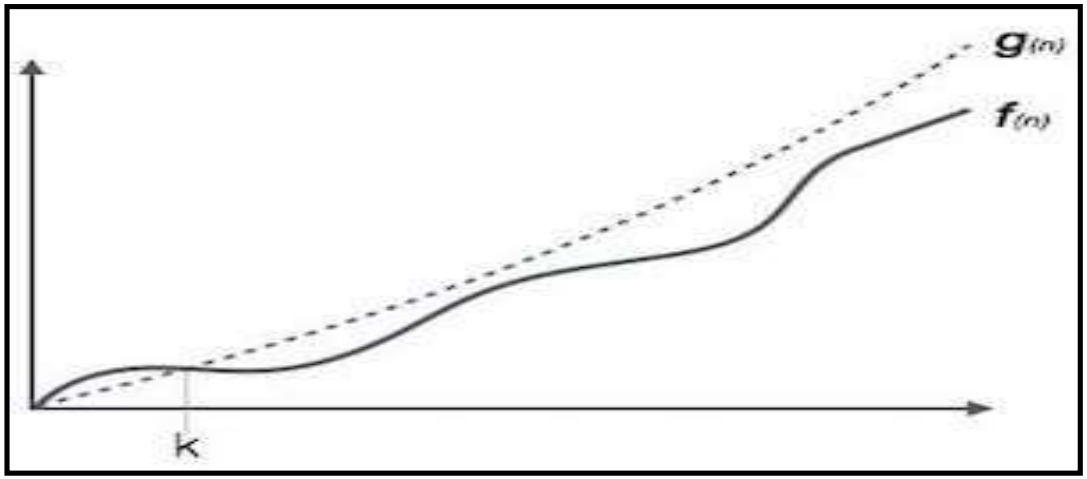
Big-O Notation

Definition: ***f*(*n*)** is in **O(*g*(*n*))** if **order of growth** of ***f*(*n*)** ≤ **order of growth** of ***g*(*n*)** (within constant multiple),

Examples:

- $10n$ is $O(n^2)$
- $5n+20$ is $O(n)$

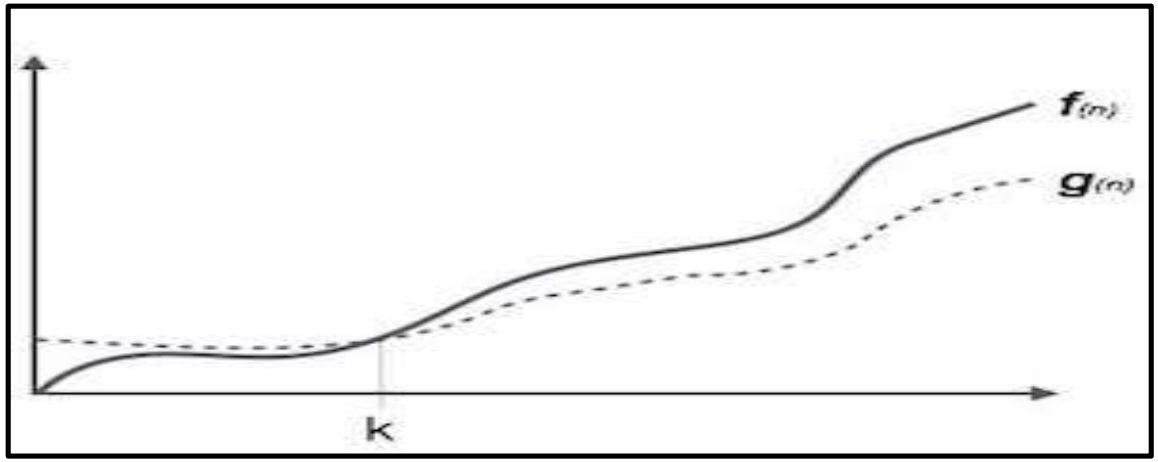
$$O(g(n)) = \left\{ \begin{array}{c} f(n) : \text{there exist positive constants } c \text{ and } n_0 \text{ s.t.} \\ 0 \le f(n) \le cg(n) \text{ for all } n \ge n_0 \end{array} \right\}$$



Omega Notation, Ω

The notation **O(n)** is the **formal way** **express**to the **upper bound** of an **algorithm's running time**.

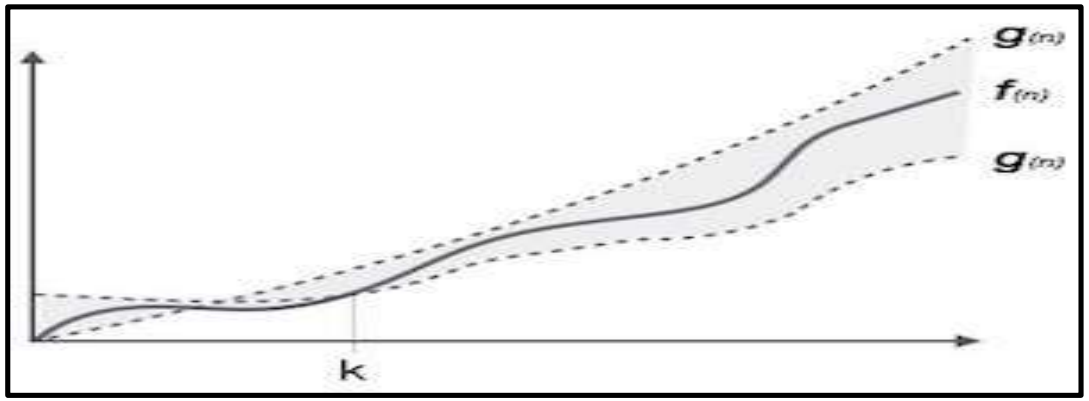
$$\Omega(g(n)) = \left\{ \begin{array}{c} f(n) : \text{there exist positive constants } c \text{ and } n_0 \text{ s.t.} \\ 0 \le cg(n) \le f(n) \text{ for all } n \ge n_0 \end{array} \right\}$$



Theta Notation, θ

The notation **θ(n)** is the **formal way** to express both the **lower bound** and the **upper bound** of an **algorithm's running time**. It is represented as follows

$$\Theta(g(n)) = \left\{ \begin{array}{c} f(n) : \text{there exist positive constants } c_1, c_2, \text{and } n_0 \text{ s.t.} \\ 0 \le c_1 g(n) \le f(n) \le c_2 g(n) \text{ for all } n \ge n_0 \end{array} \right\}$$



Common Asymptotic Notations

Common order-of-growth classifications

order of growth	name	typical code framework	description	example	$T(2N) / T(N)$
1	**constant**	`a = b + c;`	statement	add two numbers	1
$\log N$	**logarithmic**	`while (N > 1)` `{ N = N / 2; ... }`	divide in half	binary search	~ 1
N	**linear**	`for (int i = 0; i < N; i++)` `{ ... }`	loop	find the maximum	2
$N \log N$	**linearithmic**	[see mergesort lecture]	divide and conquer	mergesort	~ 2

N^2	**quadratic**	`for (int i = 0; i < N; i++)`
		`for (int j = 0; j < N; j++)`
		`{ ... }`
			double loop	check all pairs	4

N^3	**cubic**	`for (int i = 0; i < N; i++)`
		`for (int j = 0; j < N; j++)`
		`for (int k = 0; k < N; k++)`
		`{ ... }`
			triple loop	check all triples	8

2^N	**exponential**	[see combinatorial search lecture]	exhaustive search	check all subsets	$T(N)$

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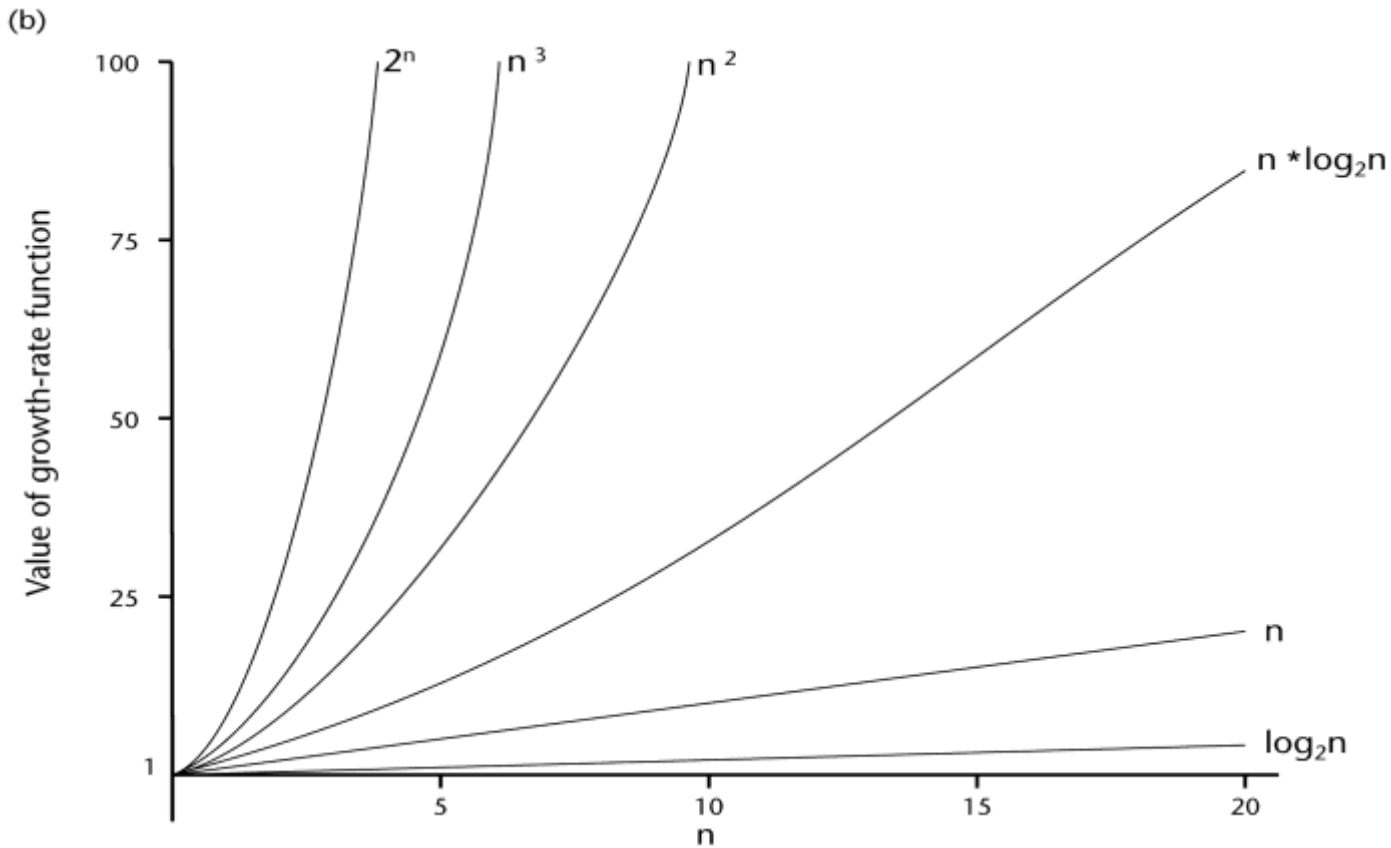
A Comparison of Growth-Rate Functions

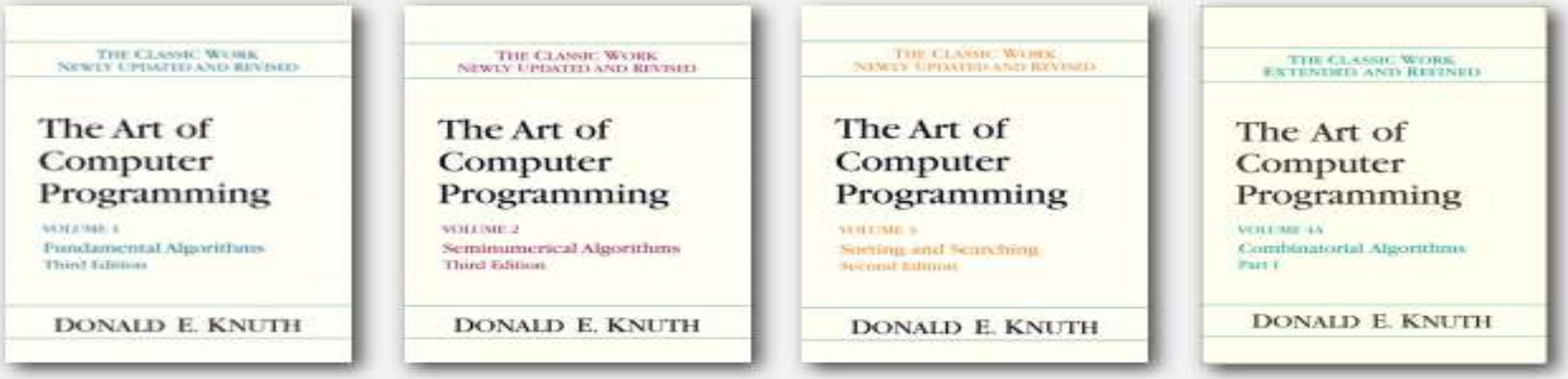


How to Compute the run time of an Algorithm

Mathematical models for running time

Total running time: sum of cost × frequency for all operations.

- Need to analyze program to determine set of operations.
- Cost depends on machine, compiler.
- Frequency depends on algorithm, input data.



In principle, accurate mathematical models are available.

How to Compute the run time of an Algorithm

Mathematical models for running time

In principle, accurate mathematical models are available.

In practice,

- Formulas can be complicated.
- Advanced mathematics might be required.
- Exact models best left for experts.

costs (depend on machine, compiler)

$$T_N = c_1 A + c_2 B + c_3 C + c_4 D + c_5 E$$

$A =$ array access
$B =$ integer add
$C =$ integer compare
$D =$ increment
$E =$ variable assignment

frequencies
(depend on algorithm, input)

Bottom line. We use approximate models in this course: $T(N) \sim c\,N^3$.

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Cost of basic operations

Cost of basic operations

Challenge. How to estimate constants.

operation	example	nanoseconds †
integer add	a + b	2.1
integer multiply	a * b	2.4
integer divide	a / b	5.4
floating-point add	a + b	4.6
floating-point multiply	a * b	4.2
floating-point divide	a / b	13.5
sine	Math.sin(theta)	91.3
arctangent	Math.atan2(y, x)	129.0
...	...	...

† Running OS X on Macbook Pro 2.2GHz with 2GB RAM

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Cost of basic operations

Cost of basic operations

Observation. Most primitive operations take constant time.

operation	example	nanoseconds †
variable declaration	`int a`	c_1
assignment statement	`a = b`	c_2
integer compare	`a < b`	c_3
array element access	`a[i]`	c_4
array length	`a.length`	c_5
1D array allocation	`new int[N]`	$c_6 N$
2D array allocation	`new int[N][N]`	$c_7 N^2$

Caveat. Non-primitive operations often take more than constant time.

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Cost of Instructions

Example: 1-SUM

Q. How many instructions as a function of input size N?

```
int count = 0;
for (int i = 0; i < N; i++)
   if (a[i] == 0)
      count++;
```

N array accesses

operation	frequency
variable declaration	2
assignment statement	2
less than compare	$N + 1$
equal to compare	N
array access	N
increment	N to $2N$

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Growth-Rate Functions – Example1

	Cost	Times
i = 1;	c1	1
sum = 0;	c2	1
while (i <= n) {	c3	n+1
i = i + 1;	c4	n
sum = sum + i;	c5	n
}		

$$T(n) = c1 + c2 + (n+1)*c3 + n*c4 + n*c5$$
$$= (c3+c4+c5)*n + (c1+c2+c3)$$
$$= a*n + b$$

➔ So, the growth-rate function for this algorithm is **O(n)**

Growth-Rate Functions – Example2

	Cost	Times
i=1;	c1	1
sum = 0;	c2	1
while (i <= n) {	c3	n+1
j=1;	c4	n
while (j <= n) {	c5	n*(n+1)
sum = sum + i;	c6	n*n
j = j + 1;	c7	n*n
}		
i = i +1;	c8	n
}		

T(n) = c1 + c2 + (n+1)*c3 + n*c4 + n*(n+1)*c5+n*n*c6+n*n*c7+n*c8

$= (c5+c6+c7)*n^2 + (c3+c4+c5+c8)*n + (c1+c2+c3)$

$= a*n^2 + b*n + c$

➔ So, the growth-rate function for this algorithm is **$O(n^2)$**

Growth-Rate Functions – Example3

	Cost	Times
for (i=1; i<=n; i++)	c1	n+1
for (j=1; j<=i; j++)	c2	$\sum_{j=1}^{n}(j+1)$
for (no=1; no<=j; no++)	c3	$\sum_{j=1}^{n}\sum_{k=1}^{j}(k+1)$
x=x+1;	c4	$\sum_{j=1}^{n}\sum_{k=1}^{j}k$

$$T(n) = c1*(n+1) + c2*\left(\sum_{j=1}^{n}(j+1)\right) + c3*\left(\sum_{j=1}^{n}\sum_{k=1}^{j}(k+1)\right) + c4*\left(\sum_{j=1}^{n}\sum_{k=1}^{j}k\right)$$

$$= a*n^3 + b*n^2 + c*n + d$$

➔ So, the growth-rate function for this algorithm is **$O(n^3)$**

Some Well-known Computational Problems

- Searching and Sorting
- Combinatorial problems
- Geometrical Problems
- Traveling salesman problem
- Knapsack problem
- Chess
- Towers of Hanoi
- Graph Problems

Algorithms Examples

Search Problems

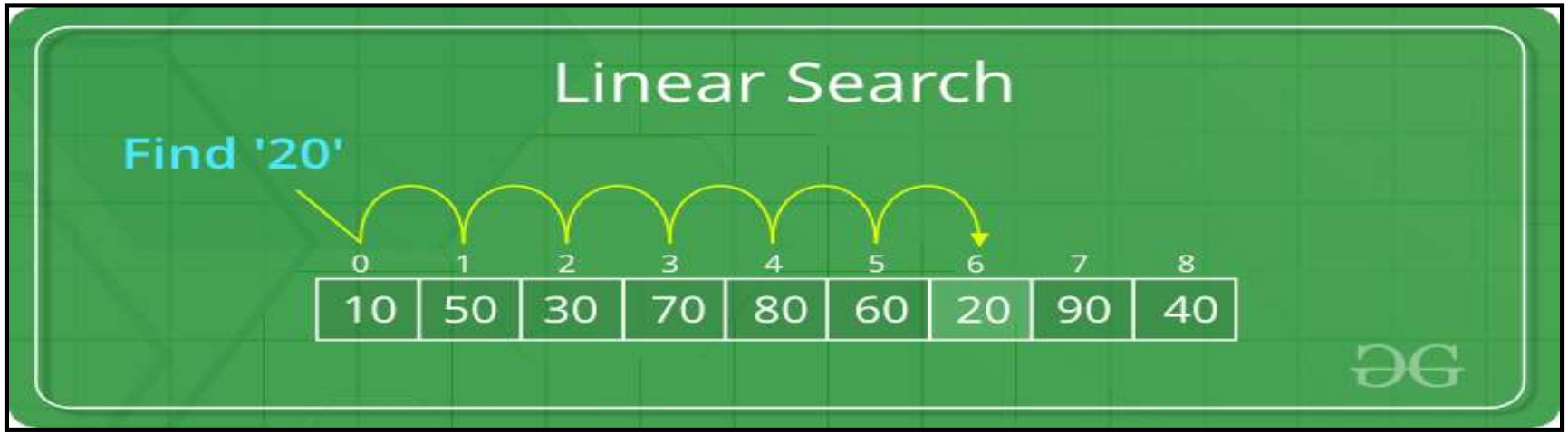
- **Statement of problem**:
  - *Input:* A sequence of *n* numbers <a_1, a_2, …, a_n>
  - Key= item to be Searched
  - *Output: Index of the item to be searched*
- **Instance**: The sequence <5, 3, 2, 8, 3>

- **Algorithms**:
  - Sequential Search
  - Binary Search(for sorted arrays)

Linear Search

Worst and average case time complexity is order of N. So we need a better algorithm.

Key=20



average case time complexity for serial search is O(n).

The worst case time for serial search requires *n* array accesses: O(*n*)

Binary Search

Goal. Given a sorted array and a key, find index of the key in the array?

Binary search. Compare key against middle entry.

- Too small, go left.
- Too big, go right.
- Equal, found.

successful search for 33

6	13	14	25	33	43	51	53	64	72	84	93	95	96	97
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
lo														hi

Activa

Binary Search

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0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
lo														hi

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Binary Search Implementation

Binary search: Java implementation

Trivial to implement?

- First binary search published in 1946.
- First bug-free one in 1962.
- Bug in Java's `Arrays.binarySearch()` discovered in 2006.

```
public static int binarySearch(int[] a, int key)
{
   int lo = 0, hi = a.length-1;
   while (lo <= hi)
   {
       int mid = lo + (hi - lo) / 2;
       if      (key < a[mid]) hi = mid - 1;
       else if (key > a[mid]) lo = mid + 1;
       else return mid;
   }
   return -1;
}
```

one "3-way compare"

Invariant. If key appears in the array a[], then a[lo] ≤ key ≤ a[hi].

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Analysis of Binary Search

Binary search: mathematical analysis

Proposition. Binary search uses at most $1 + \lg N$ key compares to search in a sorted array of size N.

Def. $T(N)$ = # key compares to binary search a sorted subarray of size $\leq N$.

Binary search recurrence. $T(N) \leq T(N/2) + 1$ for $N > 1$, with $T(1) = 1$.

↑ left or right half (floored division)

↑ possible to implement with one 2-way compare (instead of 3-way)

Pf sketch. [assume N is a power of 2]

$$
\begin{aligned}
T(N) &\leq T(N/2) + 1 && \text{[given]} \\
&\leq T(N/4) + 1 + 1 && \text{[apply recurrence to first term]} \\
&\leq T(N/8) + 1 + 1 + 1 && \text{[apply recurrence to first term]} \\
&\vdots \\
&\leq T(N/N) + 1 + 1 + \ldots + 1 && \text{[stop applying, } T(1) = 1 \text{]} \\
&= 1 + \lg N
\end{aligned}
$$

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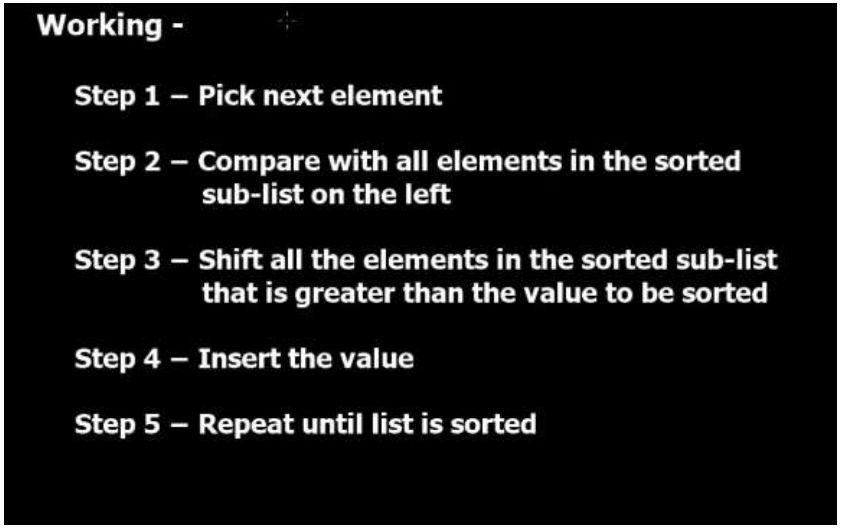
⑩**Binary search algorithm worst case O(log n) and**

⑩**best case O(1) when the central index would directly match the desired value.**

Sorting Problems

- **Statement of problem**:
  - *Input:* A sequence of n numbers $<a_1, a_2, \ldots, a_n>$
  - *Output:* A reordering of the input sequence $<a'_1, a'_2, \ldots, a'_n>$ so that $a'_i \leq a'_j$ whenever $i < j$
- **Instance**: The sequence <5, 3, 2, 8, 3>
- **Algorithms**:
  - Selection sort
  - Insertion sort
  - **Merge sort**
  - **Quick Sort**
  - (many others)

- Selection sort
- Given an array of length n,
  - Search elements 0 through n-1 and select the smallest
    - Swap it with the element in location 0
  - Search elements 1 through n-1 and select the smallest
    - Swap it with the element in location 1
  - Search elements 2 through n-1 and select the smallest
    - Swap it with the element in location 2
  - Search elements 3 through n-1 and select the smallest
    - Swap it with the element in location 3
  - Continue in this fashion until there's nothing left to search
- Insertion sort



String Processing

- **A string** is a sequence of characters from an alphabet.
- **Text strings:** letters, numbers, and special characters.
- **String matching**: searching for a given word/pattern in a **text**.

Examples:

(i) searching for a word or phrase on WWW or **in a Word document**

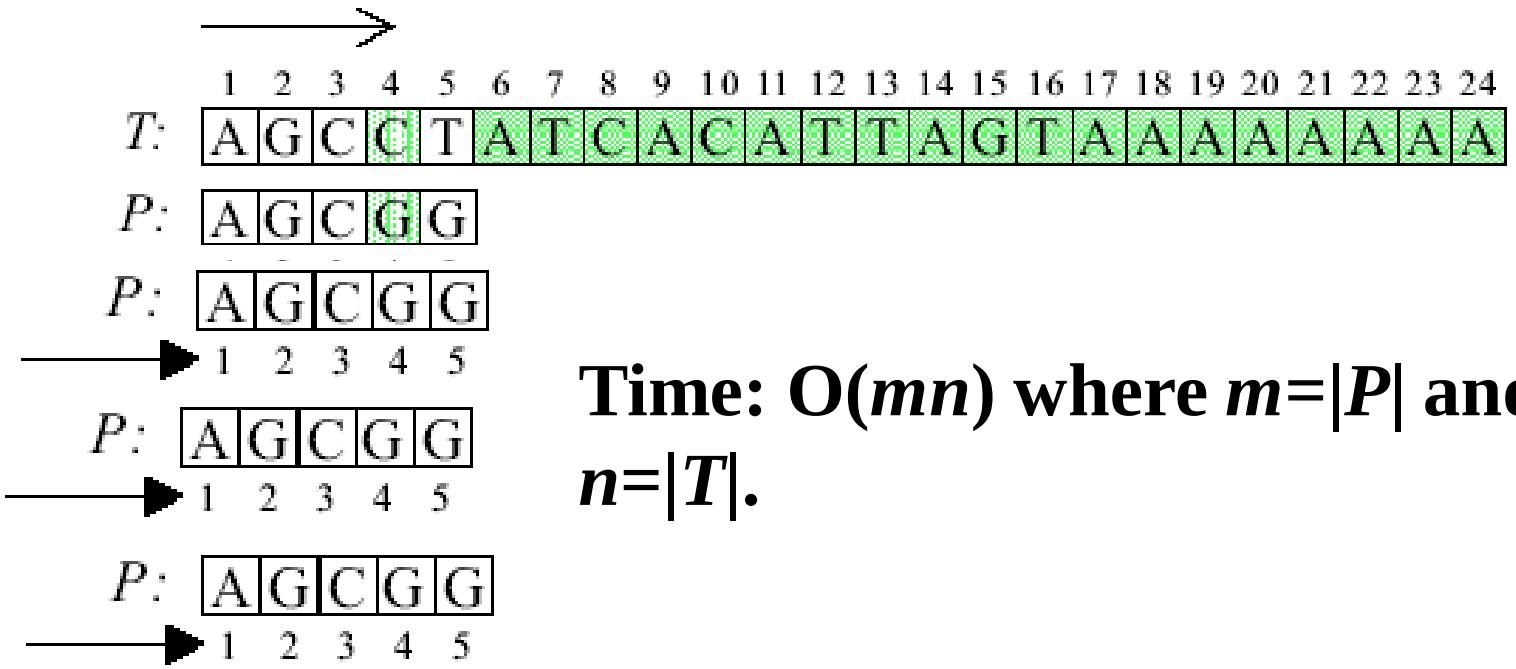
(ii) searching for a short read in the reference genomic sequence

String Matching

- Given a text string T of length n and a pattern string P of length m, the exact string matching problem is to find all occurrences of P in T.
- Example: T="AGCTTGA" P="GCT"
- **Applications**:
  - Searching keywords in a file
  - Searching engines (like Google and Openfind)
  - Database searching (GenBank)
- More string matching algorithms (with source codes): http://www-igm.univ-mlv.fr/~lecroq/string/

String Matching

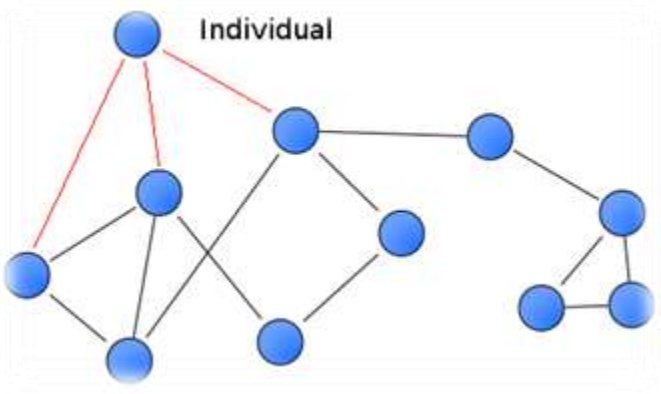
- **Brute Force algorithm**
  - The brute force algorithm consists in checking, **at all positions in the text between 0** and ***n-m***, whether an occurrence of the pattern starts there or not. Then, after each attempt, **it shifts the pattern by exactly one position to the right**.



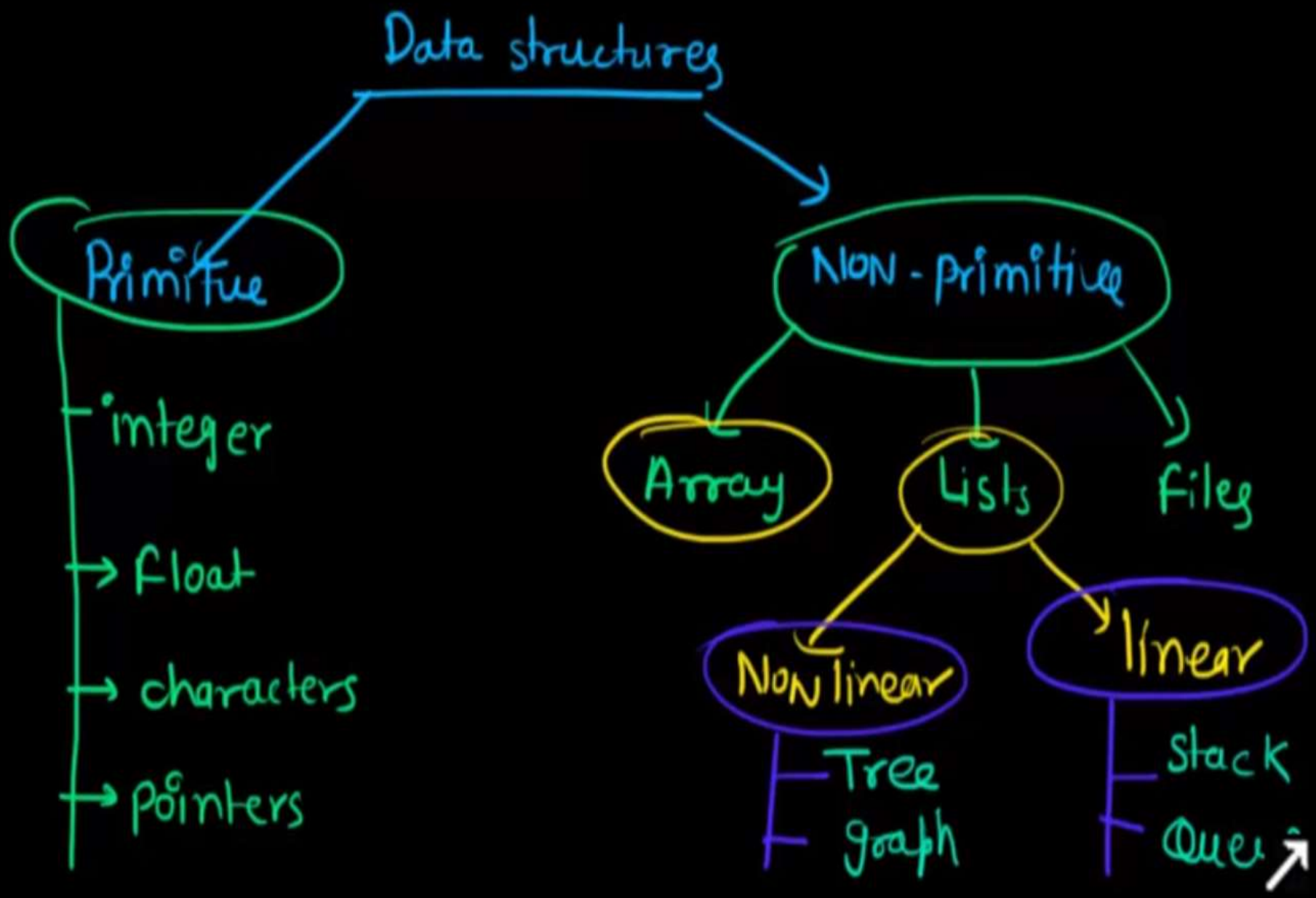
Time: O(*mn*) where *m*=|*P*| and *n*=|*T*|.

Graph Problems

- **Informal definition**
  - **A graph** is a collection of **points** called **vertices**, some of which are **connected by line segments** called **edges**.
- **Modeling Real-life Problems**
  - Modeling WWW
  - Communication networks
  - Project scheduling …
- **Examples of Graph Algorithms**
  - Graph traversal algorithms
  - Shortest-path algorithms
  - ……..



Data Structures Review


Data structures
Primitive
integer
float
characters
pointers
Non-primitive
Array
Lists
Files
Non linear
Tree
graph
linear
Stack
Queue

Fundamental data structures

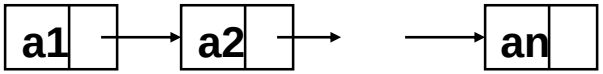
- **list**
  - array
  - linked list
  - string
- **stack**
- **queue**
- **priority queue**

♌ **graph**

♌ **tree and binary tree**

Linear Data Structures

- **Arrays**
  - **A sequence** of **n** items of the **same data type** that are **stored contiguously in computer memory** and **made accessible** by specifying a value of the **array's index**.
- **Linked List**
  - **A sequence** of **zero** or **more nodes** each containing **two kinds of information**: **some data** and **one or more links** called **pointers** to other nodes of the linked list.
  - Singly linked list (next pointer)
  - Doubly linked list (next + previous pointers)



- **Arrays**
  - **fixed length (need preliminary reservation of memory)**
  - **contiguous memory locations**
  - **direct access**
  - **Insert/delete**
- **Linked Lists**
  - **dynamic length**
  - **arbitrary memory locations**
  - **access by following links**
  - **Insert/delete**

Stacks and Queues

- **Stacks**
  - A stack of plates
    - **insertion/deletion** can be done **only at the top**.
    - LIFO
  - Two operations (**push** and **pop**)
- **Queues**
  - A queue of customers waiting for services
    - **Insertion/enqueue** from the **rear** & **deletion**/**dequeue** from the **front**.
    - FIFO
  - **Two operations** (**enqueue** and **dequeue**)

Priority Queue and Heap

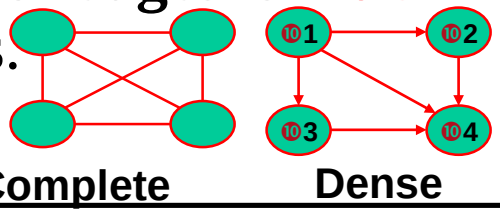
- **Priority queues** (implemented using **heaps**)
  - **A data structure** for maintaining a set of elements, each associated **with a key/priority**, with the following operations:
    - **Finding** the element with the **highest priority**
    - **Deleting** the element with the **highest priority**
    - **Inserting** a new element
  - **Scheduling jobs on a shared computer**

Graphs

- **Formal definition**
  - A graph $G = <V, E>$ is defined by a pair of two sets: a **finite set V** of items called **vertices** and a set E of vertex pairs called **edges**.
- **Undirected** and **directed** graphs (**digraphs**).

Complete, **dense**, and **sparse** graphs

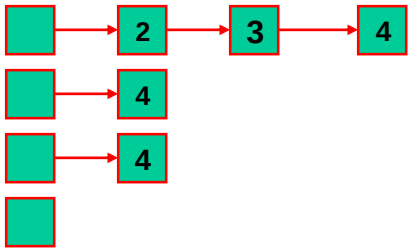
- **A graph** with every pair of its vertices connected by an **edge** is called **complete**, $K_{|V|}$
- **In Dense graph,** the number of edges is close to the maximal **number of edges**.
- A **sparse** graph in which the **number of edges** is **much less than** the possible number of edges.



Graph Representation

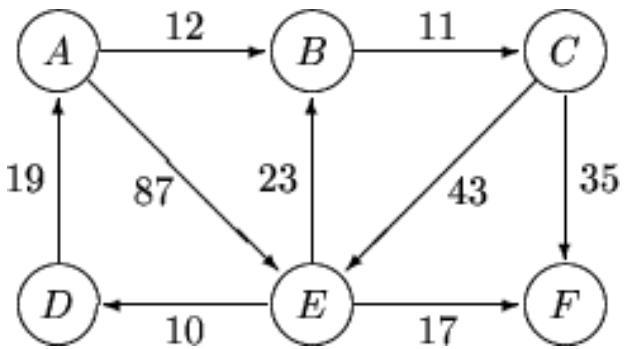
- **Adjacency matrix O(n^2) ----> better for dense graph**
  - n x n boolean matrix if |V| is n.
  - The **element** on the **ith** row and **jth** column is **1** if there's an edge from ith vertex to the jth vertex; otherwise 0.
  - The **adjacency matrix** of an undirected graph is **symmetric**.
- **Adjacency linked lists O(n+2e)--->better for sparce graph**
  - **A collection of linked lists**, one for each vertex, that contain all the vertices adjacent to the list's vertex.
- **Which data structure would you use if the graph is a 100-node star shape?**

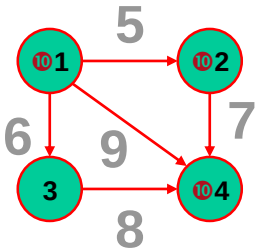
$$\begin{matrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{matrix}$$



Weighted Graphs

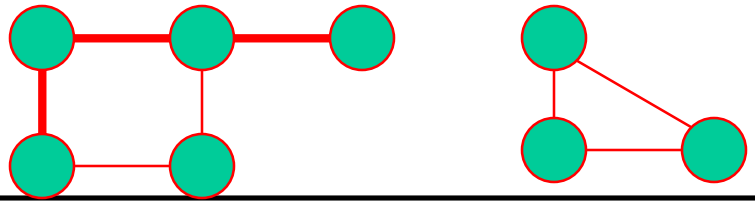
- **Graphs** or **digraphs** with numbers assigned to the **edges**. it may be called **a weighted graph**.





Graph Properties -- Paths and Connectivity

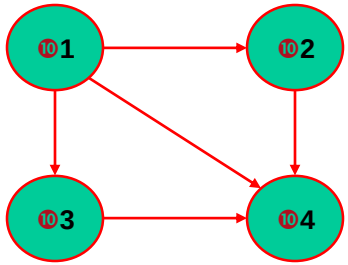
- **Paths**
  - A **path** from vertex *u* to *v* of a graph G is defined **as a sequence of adjacent** (connected by an edge) **vertices** that **starts** with ***u*** and **ends** with ***v***.
  - **Simple paths**: a path in a graph which **does not have repeating vertices..**
  - **Path lengths**: the **number of edges**, or **the number of vertices** – **1**.
- **Connected graphs**
  - A graph is said to be **connected** if for **every pair of its vertices** **u** and **v** there is **a path from u** to **v**.
- **Connected component**
  - The **maximum connected subgraph** of a given graph.



Graph Properties – A cyclicity

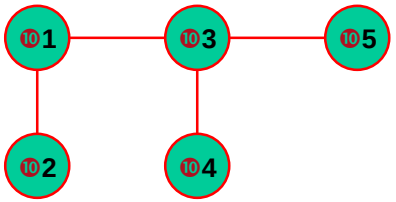
- **Cycle**
  - A **simple path** of a **positive length** that **starts** and **ends** a the **same vertex**.

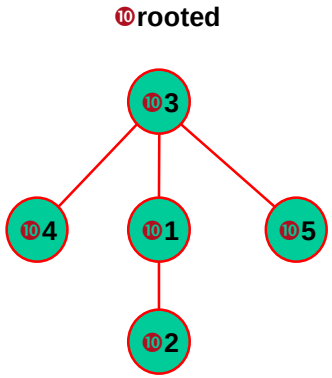
- **Acyclic graph**
  - A graph without cycles
  - DAG (Directed Acyclic Graph)

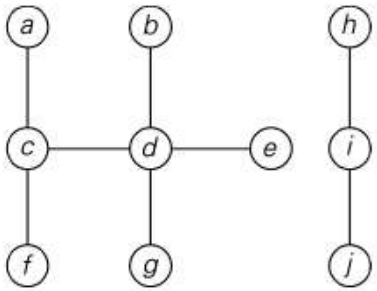


Trees

- **Trees**
  - A **Tree** (or free tree) is a connected **acyclic graph**.
  - **Forest**: a graph that has **no cycles** but is **not necessarily connected**.
- **Properties of trees**
  - For **every two vertices** in a **tree** there **always exists exactly** one simple path from one of these vertices to the other.
    - **Rooted trees**: The above property makes it possible to select an **arbitrary vertex** in a **free tree** and consider it as the root of the so called rooted tree.
    - Levels in a rooted tree.



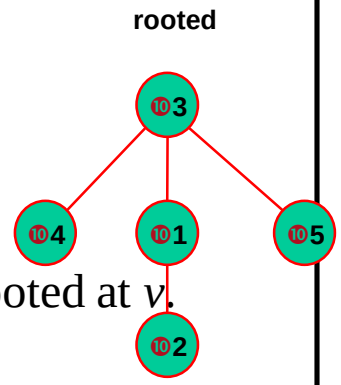




Forest

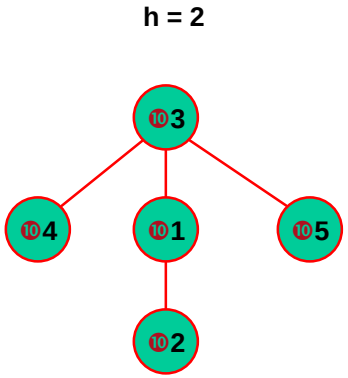
Rooted Trees (I)

- **Ancestors/predecessors**
  - For **any vertex *v*** in a tree ***T***, **all the vertices** on the simple **path** from the **root** to **that vertex** are called **ancestors**.
- **Descendants/children**
  - **All the vertices** for which a **vertex *v*** is an **ancestor** are said to be descendants of ***v***.
- **Parent**, **child** and **siblings**
  - If *(u, v)* is the last edge of the simple path from the root to vertex *v*, *u* is said to be the parent of *v* and *v* is called a child of *u*.
  - **Vertices** that have the **same parent** are called **siblings**.
- **Leaves**
  - **A vertex** without **children** is called a **leaf**.
- **Subtree**
  - **A vertex *v*** with **all its descendants** is called the subtree of *T* rooted at *v*.



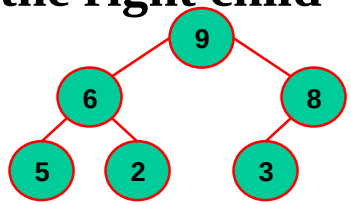
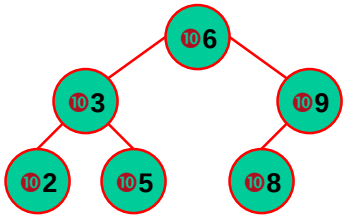
Rooted Trees (II)

- **Depth of a vertex**
  - The **length of the simple path** from the **root** to the vertex.
- **Height of a tree**
  - The **length of the longest** simple **path** from the **root** to a **leaf**.



Ordered Trees

- **Ordered trees**
  - it is **a rooted tree** in which all the children of **each vertex** are **ordered**.
- **Binary trees**
  - it is a **tree data structure** in which **each node** has **at most two children**, which are referred to as the **left child** and **the right child**
- **Binary search trees**
  - Each **vertex is assigned a number**.



  - **A number assigned** to each **parental vertex** is **larger** than all the numbers in **its left subtree** and **smaller than** all the numbers in its right subtree.